Module 1: Geometric Primitives & Operations

Module 2: Differential Geometry & Curvature

Module 3: Field Equations, Constraints & ADM (3+1)

Module 4: Gauge, Diffeomorphism & Invariant Structure

Module 5: Matter Models & Physical Constraints

Module 6: Controlled Perturbation Mapping

N1: Coordinates
x^μ

N2: Metric Tensor
g_μν

N3: Inverse Metric
g^μν

Inverse matrix

N4: Metric Determinant
g

Determinant

N5: Kronecker Delta
δ^μ_ν

Trace g_μν g^μν = 4

N6: Coordinate Transform
x'^μ = f^μ(x)

Tensor passive transform

N7: Index operations
Raising/Lowering

Musical isomorphism N8: Symmetrization

A_(μν)

Symmetrization

N9: Antisymmetrization
A_[μν]

Antisymmetrization

N14: Christoffel Symbol
Γ^ρ_μν

Levi-Civita connection

N19: Einstein Tensor
G_μν

Einstein tensor definition

N24: Stress-Energy Tensor
T_μν

Perfect fluid stress-energy N31: Foliation / Time Function

t (Σ_t)

Foliation variables (n, γ, N, N^i) N54: Metric Perturbation

h_μν = g_μν - ḡ_μν

Metric perturbation setup

N58: Diffeo Equivalence Class
[g]

Equivalence class definition N61: Transformed Active Field

φ* T

Active field pullback

N62: Killing Vector Field
ℒ_ξ g = 0 (Isometry)

Killing equation constraint

N72: Fluid Rest-Space Projector
h_μν = g_μν + u_μ u_ν

Rest-space projector N77: Linearised Christoffel

δΓ^ρ_μν

Linearised Christoffel definition

Musical isomorphism Trace g_μν g^μν = 4 Tensor passive transform Levi-Civita connection

N16: Ricci Tensor
R_μν

Contraction of Ricci

Density passive transform

N10: Volume Form
ε_μνρσ

Volume form definition

N11: Tetrad (Vierbein)
e^a_μ

e^a_μ e^b_ν η_ab = g_μν

N12: Partial Derivative
∂_μ

Levi-Civita connection

N15: Riemann Curvature
R^ρ_σμν

Riemann tensor definition

N13: Covariant Derivative
∇_μ

First Bianchi Identity

Second Bianchi Identity

N22: Commutator of ∇
[∇_μ, ∇_ν]

Commutator definition

Riemann tensor definition

N20: Geodesic Equation
∇_v v = 0 (ODE)

Geodesic ODE autoparallels

Contraction of Riemann

N18: Weyl Tensor (4D)
C^ρ_σμν

Weyl decomposition

N23: Kretschmann Scalar
K = R^μνρσ R_μνρσ

Quadratic contraction

N17: Ricci Scalar
R

Contraction of Ricci

Weyl decomposition Einstein tensor definition

Weyl decomposition Einstein tensor definition

Algebraic recoverability (4D) N26: Einstein Field Eq

G_μν = κ T_μν

Einstein Field Equation

N21: Bianchi Identities
1st & 2nd

N27: Stress-Energy Conservation
∇_μ T^μν = 0

∇_μ G^μν = 0 →∇_μ T^μν = 0

N82: Linearised Conservation
∇̄_μδT^μν = 0

Linearised conservation identity

Commutator definition

Einstein Field Equation

N28: Projected Energy Density
ρ = n^μ n^ν T_μν

Stress-energy projections

N29: Projected Momentum Density
S_i = -γ_i^μ n^ν T_μν

Stress-energy projections

N30: Projected Spatial Stress
S_ij = γ_i^μγ_j^ν T_μν

Stress-energy projections

N64: Fluid Energy Density
ρ

Projections of perfect fluid T

N65: Isotropic Pressure
p

Projections of perfect fluid T

N73: Weak Energy Condition
T_μν t^μ t^ν ≥ 0 (WEC)

Weak energy condition inequality

N74: Null Energy Condition
T_μν k^μ k^ν ≥ 0 (NEC)

Null energy condition inequality

N75: Dominant Energy Condition
WEC + non-spacelike flux (DEC)

Dominant energy condition inequality

N76: Strong Energy Condition
SEC

Strong energy condition inequality

N25: Einstein Grav Constant
κ = 8πG/c^4

Einstein Field Equation

∇_μ G^μν = 0 → ∇_μ T^μν = 0

N41: Hamiltonian Constraint
⅋ = R^(3) + K^2 - K_ij K^ij - 2κρ = 0

EFE ↔ ADM Equivalence

N42: Momentum Constraint
ℳ^i = D_j(K^ij - γ^ij K) - κ S^i = 0

EFE ↔ ADM Equivalence

N43: Kinematic Evolution
∂_t γ_ij = -2N K_ij + D_i N_j + D_j N_i

EFE ↔ ADM Equivalence

N44: Dynamical Evolution of K
∂_t K_ij

EFE ↔ ADM Equivalence

N63: Fluid 4-Velocity
u^μ

Euler & energy conservation Euler & energy conservation

Euler & energy conservation

N67: Scalar Field
φ

Klein-Gordon equation of motion

N68: Scalar Potential
V(φ)

Klein-Gordon equation of motion

N69: EM Field Tensor
F_μν = ∂_μ A_ν - ∂_ν A_μ

Maxwell equations of motion

N70: 4-Current Density
J^μ

Maxwell equations of motion

Recoverability of T_μν

Hamiltonian constraint Dynamical evolution of K

Recoverability of T_μν

Momentum constraint

Recoverability of T_μν

Dynamical evolution of K

N45: Trace of Spatial Stress
S = γ^ij S_ij

Trace of spatial stress

N32: Unit Normal Vector
n^μ

Normal vector to slices

N33: Projector onto Slice
γ^μ_ν = δ^μ_ν + n^μ n_ν

Projector onto slice

N34: Induced Metric (3-metric)
γ_ij

Induced metric on slice

N35: Lapse Function
N

Lapse function

N36: Shift Vector
N^i

Shift vector

Stress-energy projections

Stress-energy projections

Lapse/Shift definition of normal

N38: 3D Covariant Derivative
D_i

3D covariant geometry

N39: 3D Christoffel Symbol
Γ^(3)i_jk

3D covariant geometry

N40: 3D Curvature Tensor

R^(3)_ijkl

3D covariant geometry

Kinematic evolution Dynamical evolution of K Trace of spatial stress

N46: ADM Line Element
ds^2 = -N^2 dt^2 + γ_ij(dx^i + N^i dt)(dx^j + N^j dt)

ADM line element representation

N47: Gauss-Codazzi Relations
Hypersurface geometry

Gauss-Codazzi relations

Lapse/Shift definition of normal

Kinematic evolution Dynamical evolution of K

ADM line element representation Lapse/Shift definition of normal

Kinematic evolution Dynamical evolution of K

ADM line element representation

N37: Extrinsic Curvature
K_ij

Kinematic evolution Momentum constraint Gauss-Codazzi relations Hamiltonian constraint Dynamical evolution of K

Momentum constraint

Hamiltonian constraint Dynamical evolution of K

N48: Constraint Propagation
Theorem

Constraint preservation Constraint preservation Constraint preservation Constraint preservation

N49: Active Diffeo
φ: M → M

N52: Lie Derivative

ℒ_ξ

Active field pullback Lie derivative definition

N50: Pullback / Pushforward
φ*, φ_*

N51: Infinitesimal Generator
ξ^μ

Isometry generator test Lie derivative definition

N60: Coordinate Scalar
R, K, ... (passive invariant)

Active scalar shift ℒ_ξ f

N53: Background Metric
ḡ_μν

Metric perturbation setup

N56: Gauge Transform of h
δ_ξ h_μν = ∇_μξ_ν + ∇_νξ_μ

Linearised gauge transform Linearised Christoffel definition

Linearised gauge transform

N59: Perturbative Equiv Class
[h]

Perturbative equivalence class

Linearised Christoffel definition

N83: Gauge-Invariant Metric Pert
Δh_μν

Gauge-invariant combination

N55: Perturbative Gauge Generator
ξ^μ_(1)

Linearised gauge transform

Gauge-invariant combination

Gauge dependence of curvature perturbations Perturbative equivalence class

N57: Gauge-Invariant Tensor
ΔQ_μ... = δQ_μ... - ℒ_ξ Q̄

_μ...

Equivalence class definition Active scalar shift ℒ_ξ f

Perfect fluid stress-energy Perfect fluid stress-energy Rest-space projector

N66: Equation of State
p = p(ρ, s) or p = wρ

Thermodynamic closure

Perfect fluid stress-energy Thermodynamic closure

Scalar field stress-energy Scalar field stress-energy Maxwell stress-energy

N71: Cosmological Constant
Λ

Cosmological constant vacuum energy

N78: Linearised Ricci
δR_μν

Linearised Ricci definition

N79: Linearised Einstein
δG_μν

Linearised Einstein definition

N80: Linearised EFE
δG_μν = κ δT_μν

Linearised EFE field equation

Linearised conservation identity

Gauge-fixed EFE reduced PDE system

N81: Linearised Fluid Perturbations
δρ, δp, δu^μ

Linearised EFE field equation